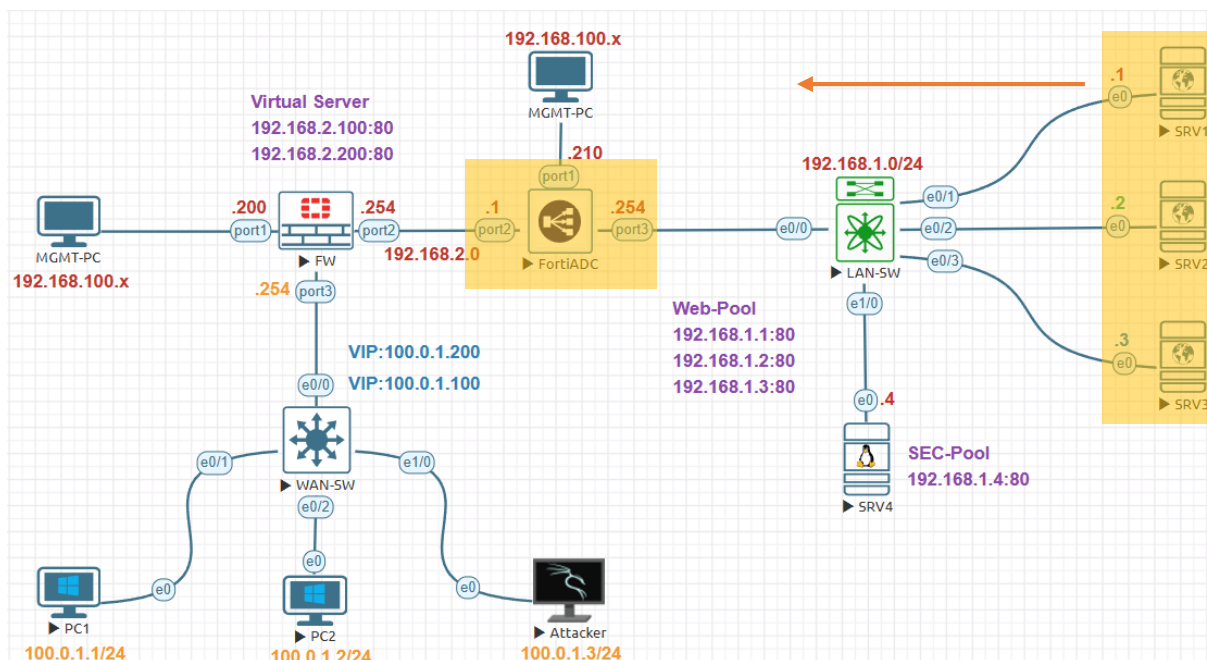


1-to-1 NAT for Servers Lab:



Management Subnet	192.168.100.0/24
FortiGate Management IP	192.168.100.200
FortiADC Management IP	192.168.100.210
Internal Servers Subnet	192.168.1.0/24
FortiADC External Subnet	192.168.2.0/24
FortiGate Firewall External	100.0.1.0/24
FortiADC Internal IP Address	192.168.1.254
FortiADC External IP Address	192.168.2.1
FortiGate Firewall Internal IP	192.168.2.254
FortiGate Firewall External IP	100.0.1.254
SRV1 IP Address	192.168.1.1
SRV2 IP Address	192.168.1.2
SRV3 IP Address	192.168.1.3
SRV3 IP Address	192.168.1.4
Virtual Sever-1	192.168.2.100
Virtual Sever-2	192.168.2.200
FortiGate VIP-1	100.0.1.100
FortiGate VIP-2	100.0.1.200
External PC1 IP Address	100.0.1.1
External PC2 IP Address	100.0.1.2
External PC3 IP Address	100.0.1.3

Go to **Networking > NAT** The configuration page displays the **1-to-1 NAT** Tab. Click **Create New** to display the configuration editor.

The screenshot shows the FortiGate web interface. On the left sidebar, the 'Network' menu is expanded, and 'NAT' is selected. The main content area displays the '1-to-1 NAT' configuration page. At the top, there is a 'Source' tab and a '1-to-1 NAT' sub-tab. Below this, there are buttons for 'Delete', '+ Create New', and '+ Add Filter'. A table with columns 'Name', 'External Interface', and 'External IP Address' is visible. Below the table, it says 'Showing 0 to 0 of 0 entries 0 rows selected Show 25 entries'.

Complete the configuration and click **Save** the configuration.

The screenshot shows the '1-to-1 NAT' configuration editor. The left sidebar is the same as the previous screenshot. The main content area shows the configuration fields for a new NAT rule. The fields are: Name (One-to-One-NAT), External Interface (port2), External Address Range (192.168.2.80), To (192.168.2.81), Mapped Address Range (192.168.1.0), To (192.168.1.1), Port Forwarding (disabled), Traffic Group (default), and Status (enabled). A red arrow points to the 'Save' button at the bottom right.

Finally, **1-to-1 NAT** rule has been created shown below.

The screenshot shows the '1-to-1 NAT' configuration page after the rule has been created. The left sidebar is the same. The main content area shows the '1-to-1 NAT' tab. The table now has one entry: 'One-to-One-NAT' with 'port2' as the External Interface and '192.168.2.80' as the External IP Address. Below the table, it says 'Showing 1 to 1 of 1 entries 0 rows selected Show 25 entries'.

Let's try to access internet again from SRV1 its reachable.

```
FortiADC SRV1 SRV2
root@SRV1:~#
root@SRV1:~# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=66.5 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=65.6 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=67.0 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=115 time=65.8 ms
^C
```

Create Virtual IPs in Firewall under **Policy & Objects>Virtual IPs**

New Virtual IP

VIP type IPv4

Name SRV1-VIP-01

Comments Write a comment... 0/255

Color Change

Network

Interface ☐ any

Type Static NAT FQDN

External IP address/range 100.0.1.80

Map to

IPv4 address/range 192.168.2.80

+ Create New EditCloneDelete Search Q					
Name	Details	Interfaces	Services	Ref.	
IPv4 Virtual IP 3					
WEB-VIP-01	100.0.1.100 → 192.168.2.100	☐ any		1	10
WEB-VIP-02	100.0.1.200 → 192.168.2.200	☐ any		1	0
SRV1-VIP-01	100.0.1.80 → 192.168.2.80	☐ any		1	44

Create Firewall Policy to allow the traffic from outside to SRV1 under the **Policy & Objects>Firewall Policy**

Dashboard

Network

Policy & Objects

Firewall Policy

IPv4 DoS Policy

Addresses

Internet Service Database

Services

Schedules

Virtual IPs

IP Pools

Protocol Options

Traffic Shaping

Security Profiles

VPN

User & Authentication

System

Edit Policy

Name

WAN-to-SRV1

Incoming Interface

WAN (port3)

Outgoing Interface

FortiADC-LAN (port2)

Source

all

Destination

SRV1-VIP-01

Schedule

always

Service

ALL

Action

ACCEPT

DENY

Inspection Mode

Flow-based

Proxy-based

Firewall / Network Options

NAT

Protocol Options

default

+ Create New

Edit

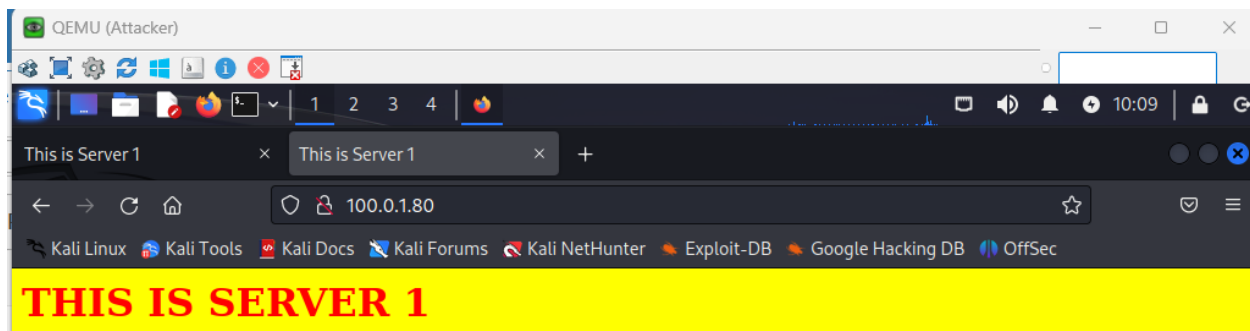
Delete

Policy Lookup

Search

Name	From	To	Source	Destination	Schedule	Service
FortiADC-to-Internet	FortiADC-LAN (port2)	MGMT (port1)	all	all	always	ALL
WAN-to-FortiADC	WAN (port3)	FortiADC-LAN (port2)	all	WEB-VIP-01 WEB-VIP-02	always	ALL
WAN-to-SRV1	WAN (port3)	FortiADC-LAN (port2)	all	SRV1-VIP-01	always	ALL
WAN-to-Internet	WAN (port3)	MGMT (port1)	all	all	always	ALL
Implicit Deny	any	any	all	all	always	ALL

On the Kali Linux-External PC, open a browser, and then connect to the virtual server at <http://100.0.1.80>



Open Console or SSH connection to the FortiADC, Enter the following CLI command to enable the sniffer and capture the traffic:

FortiADC # diagnose sniffer packet any "port 80" 4

```
FortiADC # diagnose sniffer packet any " port 80 " 4
interfaces=[any]
filters=[ port 80 ]
5.564308 port2 in 100.0.1.3.45998 -> 192.168.2.80.80: syn 3668461389
5.564418 port3 out 100.0.1.3.45998 -> 192.168.1.1.80: syn 3668461389
5.565294 port3 in 192.168.1.1.80 -> 100.0.1.3.45998: syn 4103995500 a
5.565309 port2 out 192.168.2.80.80 -> 100.0.1.3.45998: syn 4103995500
5.566237 port2 in 100.0.1.3.45998 -> 192.168.2.80.80: ack 4103995501
5.566246 port3 out 100.0.1.3.45998 -> 192.168.1.1.80: ack 4103995501
5.566271 port2 in 100.0.1.3.45998 -> 192.168.2.80.80: psh 3668461390
5.566276 port3 out 100.0.1.3.45998 -> 192.168.1.1.80: psh 3668461390
5.566794 port3 in 192.168.1.1.80 -> 100.0.1.3.45998: ack 3668461767
5.566802 port2 out 192.168.2.80.80 -> 100.0.1.3.45998: ack 3668461767
5.567662 port3 in 192.168.1.1.80 -> 100.0.1.3.45998: psh 4103995501 a
5.567670 port2 out 192.168.2.80.80 -> 100.0.1.3.45998: psh 4103995501
5.568404 port2 in 100.0.1.3.45998 -> 192.168.2.80.80: ack 4103995974
5.568411 port3 out 100.0.1.3.45998 -> 192.168.1.1.80: ack 4103995974
```

Port Forwarding:

Go to **Networking > NAT** The configuration page displays the **1-to-1 NAT** Tab. Edit enable **Port Forwarding** protocol **TCP** make External Port Range **8080** while Mapped Port Range **80** click **Save**.

1-to-1 NAT

Name

One-to-One-NAT

External Interface

port2

External Address Range

192.168.2.80

Example: 192.0.2.0

To

192.168.2.80

Mapped Address Range

192.168.1.1

Example: 198.51.100.0

To

192.168.1.1

Example: 198.51.100.9

Port Forwarding

☒

Protocol

☒ TCP ☐ UDP

External Port Range

8080

To

8080

Mapped Port Range

80

To

80

Traffic Group

default

Status

☒

Save

Cancel

Finally, the Port Forwarding is enabled.

Source	1-to-1 NAT			
Delete	+ Create New	Add Filter		
Name	External Interface	External IP Address	Mapped Address Range	Port Forwarding
One-to-One-NAT	port2	192.168.2.80	192.168.1.1 - 192.168.1.1	Enable

Also Edit the existing Virtual IP in Firewall enable Port Forwarding Protocol TCP Port Mapping Type one to one External Service port **8080** and Map to IPV4 port **8080**

Edit Virtual IP

VIP type

IPv4

Name

SRV1-VIP-01 ...

Comments

Write a comment... 0/255

Color

Change

Network

Interface i

any ▼

Type

Static NAT

External IP address/range i

100.0.1.80

Map to

IPv4 address/range

192.168.2.80

☐ Optional Filters

☒ Port Forwarding

Protocol

TCP
UDP
SCTP
ICMP

Port Mapping Type

One to one
Many to many

External service port i

8080

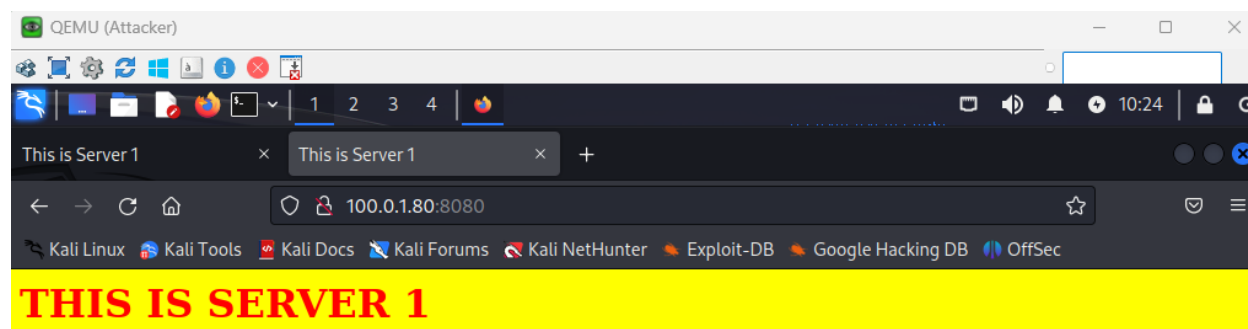
Map to IPv4 port

8080

+ Create New
 Edit
 Clone
 Delete
Search

Name ↕	Details ↕	Interfaces ↕	Services ↕
IPv4 Virtual IP 3			
WEB-VIP-01	100.0.1.100 → 192.168.2.100	☐ any	
WEB-VIP-02	100.0.1.200 → 192.168.2.200	☐ any	
SRV1-VIP-01	100.0.1.80 → 192.168.2.80 (TCP: 8080 → 8080)	☐ any	

On the Kali Linux-External PC, open a browser, and then connect to the virtual server at <http://100.0.1.80:8080>



FortiADC # diagnose sniffer packet any " port 80 or 8080 " 4

```
FortiADC # diagnose sniffer packet any " port 80 or 8080 " 4
interfaces=[any]
filters=[ port 80 or 8080 ]
5.069917 port2 in 100.0.1.3.51796 -> 192.168.2.80.8080: syn 2418110191
5.069966 port3 out 100.0.1.3.51796 -> 192.168.1.1.80: syn 2418110191
5.070786 port3 in 192.168.1.1.80 -> 100.0.1.3.51796: syn 4142548535 ack
5.070803 port2 out 192.168.2.80.8080 -> 100.0.1.3.51796: syn 4142548535
5.071975 port2 in 100.0.1.3.51796 -> 192.168.2.80.8080: ack 4142548536
5.071985 port3 out 100.0.1.3.51796 -> 192.168.1.1.80: ack 4142548536
5.074837 port2 in 100.0.1.3.51796 -> 192.168.2.80.8080: psh 2418110192
5.074852 port3 out 100.0.1.3.51796 -> 192.168.1.1.80: psh 2418110192
5.075592 port3 in 192.168.1.1.80 -> 100.0.1.3.51796: ack 2418110574
5.075601 port2 out 192.168.2.80.8080 -> 100.0.1.3.51796: ack 2418110574
5.076330 port3 in 192.168.1.1.80 -> 100.0.1.3.51796: psh 4142548536 ack
5.076340 port2 out 192.168.2.80.8080 -> 100.0.1.3.51796: psh 4142548536
5.077796 port2 in 100.0.1.3.51796 -> 192.168.2.80.8080: ack 4142549009
5.077808 port3 out 100.0.1.3.51796 -> 192.168.1.1.80: ack 4142549009
5.104670 port2 in 100.0.1.3.51796 -> 192.168.2.80.8080: psh 2418110574
5.104698 port3 out 100.0.1.3.51796 -> 192.168.1.1.80: psh 2418110574
5.105715 port3 in 192.168.1.1.80 -> 100.0.1.3.51796: psh 4142549009 ack
5.105729 port2 out 192.168.2.80.8080 -> 100.0.1.3.51796: psh 4142549009
5.147728 port2 in 100.0.1.3.51796 -> 192.168.2.80.8080: ack 4142549512
5.147750 port3 out 100.0.1.3.51796 -> 192.168.1.1.80: ack 4142549512
5.252667 port2 in 100.0.1.3.51796 -> 192.168.2.80.8080: fin 2418110911
5.252691 port3 out 100.0.1.3.51796 -> 192.168.1.1.80: fin 2418110911
5.252720 port2 in 100.0.1.3.51810 -> 192.168.2.80.8080: syn 3414149908
```